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7 Waves

An understanding of colour from Cambridge IGCSE/O Level Physics or equivalent is assumed.

7.1 Progressive waves

Candidates should be able to:

1 describe what is meant by wave motion as illustrated by vibration in ropes, springs and ripple tanks

2 understand and use the terms displacement, amplitude, phase difference, period, frequency, wavelength and speed

3 understand the use of the time-base and *y*-gain of a cathode-ray oscilloscope (CRO) to determine frequency and amplitude

4 derive, using the definitions of speed, frequency and wavelength, the wave equation $v = f\lambda$

5 recall and use $v = f\lambda$

6 understand that energy is transferred by a progressive wave

7 recall and use *intensity* = *power/area* and *intensity* $\propto$ (*amplitude*)^2 for a progressive wave

7.2 Transverse and longitudinal waves

Candidates should be able to:

1. compare transverse and longitudinal waves
2. analyse and interpret graphical representations of transverse and longitudinal waves

7.3 Doppler effect for sound waves

Candidates should be able to:

1. understand that when a source of sound waves moves relative to a stationary observer, the observed frequency is different from the source frequency (understanding of the Doppler effect for a stationary source and a moving observer is not required)
2. use the expression $f_o = f_s v / (v \pm v_s)$ for the observed frequency when a source of sound waves moves relative to a stationary observer

7.4 Electromagnetic spectrum

Candidates should be able to:

1. state that all electromagnetic waves are transverse waves that travel with the same speed c in free space
2. recall the approximate range of wavelengths in free space of the principal regions of the electromagnetic spectrum from radio waves to γ-rays
3. recall that wavelengths in the range 400–700 nm in free space are visible to the human eye

7.5 Polarisation

Candidates should be able to:

1. understand that polarisation is a phenomenon associated with transverse waves
2. recall and use Malus's law ($I = I_0 \cos^2\theta$) to calculate the intensity of a plane-polarised electromagnetic wave after transmission through a polarising filter or a series of polarising filters (calculation of the effect of a polarising filter on the intensity of an unpolarised wave is not required)